REQUIREMENT SPECIFICATION DOCUMENT

IntelliCare: An IoT-based system for household emergency situations

Project Title	IntelliCare: An IoT-based system for household emergency situations
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Project specification

HLR: The system shall provide a smart and reliable solution for detecting and responding to various household hazards, with a focus on improving safety and response time in emergency situations.

SW-1: The system shall continuously monitor the user's health metrics and track their location within the household environment.

- **SW-1.1**: The system shall utilize sensors integrated into a smart wristband capable of measuring heart rate, blood pressure, and SpO_2 levels, ensuring real-time health monitoring.
    - **ARCH**: The wristband is implemented as an embedded device integrating biomedical sensors. Sensor readings are periodically sampled and packaged into MQTT messages, which are transmitted over WiFi to the hub. Each metric is published under a structured topic for modular processing and visualization.
- **SW-1.2**: The system shall detect user movement and activity through the use of an accelerometer and gyroscope embedded within the wristband.
    - **ARCH**: Motion data is captured via accelerometer and gyroscope sensors within the wristband. Relevant motion events and raw data are published via MQTT to the hub, where higher-level activity recognition (e.g., fall detection) is performed.
- **SW-1.3**: The system shall determine the user's location within the house using passive infrared (PIR) sensors to enable accurate indoor positioning.

    - **ARCH**: Each room contains a PIR sensor acting as a node, connected via WiFi. These nodes publish motion detection events to MQTT topics. The hub subscribes to these topics and infers the user's location based on recent activity patterns. Device discovery and registration are handled via SDDP.

SW-2: The system shall be capable of detecting potential fire hazards within the household.

- **SW-2.1**: The system shall include an integrated smoke detection sensor in each room, enabling early identification of fire-related risks and supporting timely intervention.
    - **ARCH**: Smoke sensors are embedded within room sensor nodes and periodically send readings or alerts via MQTT topics. Upon detecting abnormal values, the node immediately publishes an alert message to ensure low-latency response.
- **SW-2.2**: The system shall incorporate a carbon monoxide (CO) sensor in every room to continuously monitor air quality and detect the presence of hazardous gases.
    - **ARCH**: CO sensors operate similarly to smoke sensors, continuously measuring gas concentration and publishing readings via MQTT. The hub evaluates CO levels against safety thresholds and generates alerts when necessary.
- **SW-2.3**: The system shall be equipped with a temperature sensor in each room, allowing continuous monitoring of environmental conditions and early detection of abnormal temperature changes.
    - **ARCH**: Temperature sensors periodically transmit environmental data via MQTT. The hub correlates temperature increases with other sensor data (smoke, CO) to improve fire detection accuracy.

SW-3: The system shall intelligently detect hazardous situations and initiate appropriate responses in real time.

- **SW-3.1**: The system shall detect incidents such as falls or abnormal health conditions by analyzing data from the wristband sensors. Based on the severity of the situation, the system shall either notify a caregiver or automatically contact emergency medical services. Upon arrival, the system shall utilize a smart door camera to recognize authorized emergency personnel and grant access by unlocking the door.
    - **ARCH**: Fall detection is implemented using motion data from the wristband (accelerometer/gyroscope), processed in the hub. Health

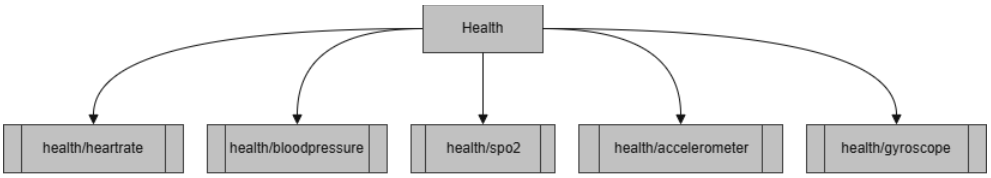
anomalies are detected by analyzing incoming vital metrics. Based on severity, the hub publishes MQTT messages to notify caregivers or trigger emergency workflows. A smart door system, connected via WiFi, receives commands to unlock upon verification from a camera module.

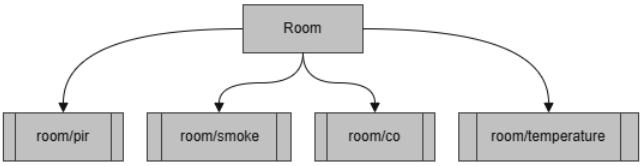
- **SW-3.2**: In the event of a fire hazard, the system shall activate appropriate safety mechanisms, including sprinkler systems and alarm signals, while simultaneously notifying firefighting services. Additionally, the system shall assist responders by guiding them to the exact location of individuals inside the house and providing controlled access through the smart door system.
    - **ARCH**: Upon fire detection, the hub publishes MQTT commands to actuator topics controlling sprinklers and alarms. Smart lighting devices, also MQTT subscribers, are activated in sequence to guide responders to the affected room. The smart door receives an unlock command, while the Qt interface displays the exact location of individuals based on sensor data.

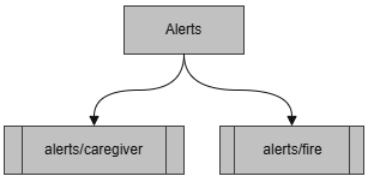
SW-4: The system shall provide a centralized dashboard for monitoring all relevant data and system activity.

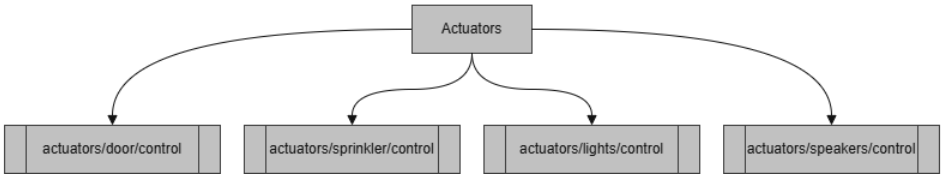
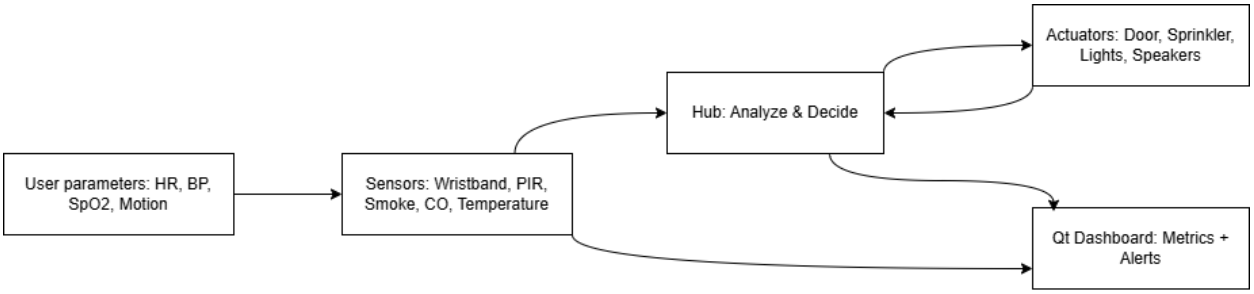
- **SW-4.1**: The dashboard shall display real-time health metrics, historical data visualizations, and sensor readings, along with an assessment of whether the user is in a normal or critical condition.
    - **ARCH**: The Qt application processes MQTT messages and renders them as live graphs and status indicators. Health metrics and environmental data are visualized using time-series components, with clear indicators for normal and critical states.
- **SW-4.2**: The system shall maintain a comprehensive log of all detected events, executed actions, and messages received from sensors, ensuring full traceability and system transparency.
    - **ARCH**: The hub logs all incoming MQTT messages, detected events, and issued commands into a persistent storage system. The Qt interface retrieves and displays these logs, enabling traceability and system diagnostics.

MQTT Topic Tree









Data Flow Demonstration